



INGERSOLL RAND: DIVERTING WOOD WASTE AT A COMPRESSOR MANUFACTURING FACILITY

SECTOR	LOCATION	FINANCIAL OVERVIEW
Industrial	Sedalia	\$187,000

BACKGROUND

Ingersoll Rand prioritizes mission-critical flow creation and industrial solutions. With more than 40 brands and approximately 15,000 employees, it provides innovative services and solutions for a broad range of industries, primarily operating in two segments: Industrial Technologies and Services and Precision and Science Technologies. In 2022, the Ingersoll Rand environmental, health and safety (EHS) team launched a project to decrease the annual amount of wood waste sent to landfill at its Sedalia, MO, compressor manufacturing facility. The wood waste accumulated at the Sedalia facility is now diverted to a mulching plant, thereby reducing landfill waste by 878,460 pounds.

All Ingersoll Rand sites have a goal of making 50% of their “in-scope sites” global manufacturing sites zero landfill by 2030. In addition, the company has a corporate-wide goal of reducing absolute water usage by 15% and Scope 1 and 2 greenhouse gas emissions by 60% from a 2020 base year. Each site is additionally expected to reduce its energy intensity by 3% each year.

SOLUTIONS

In support of Ingersoll Rand's 2030 and 2050 goals, the Sedalia EHS team surveyed staff for new ideas and received a suggestion from an employee to eliminate the practice of sending wood pallets to the local landfill. The employee suggested installing a woodchipper that would convert pallets to chips to be sent to a mulch manufacturing facility. This project changed the waste stream into a value-generating commodity and allowed the Sedalia location to make progress on its waste reduction goals and commitments. The project diverted 878,460 pounds of wood waste from local landfills, eliminated nearly 146.5 metric tons of transportation-related greenhouse gas emissions, and produced \$85,000 in savings from operational costs over the course of the 242 days the woodchipper was in operation.

To complete the project, the Sedalia team identified the appropriate chipping machine and successfully obtained approval for the required capital investment. The next step was to integrate the chipping equipment into the manufacturing process. Finally, the team negotiated a contract with a local mulching company to purchase the output from the chipper process. The completed project improved employee morale and material handling safety, saved energy and money, reduced the amount of waste sent to landfill and reduced greenhouse gas emissions.

OTHER BENEFITS

Ingersoll Rand embeds sustainability into its businesses by looking upstream into processes to find waste reduction and recycling solutions, efficient methods for reusing materials, improving energy and water efficiency, and developing take-back solutions with its suppliers. This project has been identified as a best practice and will be considered at other Ingersoll Rand facilities around the world. A similar project was implemented in Bentleyville Nash and was the inspiration for this project.

Waste Reduction Summary:

Measurement	Amount	Units
Waste Sent to Landfill		
Baseline (2021)	1,486,700	Pounds
Actual (2023)	878,460	Pounds
Reduction Amount (%):	41%	Pounds
Cost		
Disposal Cost Savings	\$16,730	\$55/ton
Cost Per Load Fee (14-ton disposal truck)	\$2,870	\$132/per trip
Annual Savings	\$19,600	

IMAGE GALLERY



WOODCHIPPER MACHINE